NEWTON SCHOOL

PRE-CALCULUS | LECTURE NOTES

Sums and the Dimension Jump

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Standard: *CCSS.MATH.CONTENT.HSF-BF.A.1 — Build a function that models a relationship between two quantities; recognize patterns in power sums and their connection to integration (calculus preview).*

Big Idea: *The sum of d-dimensional terms always produces a $(d+1)$-dimensional quantity — a pattern that secretly is calculus.*

1. Something Surprising About Sums

You have been summing integers, squares, and cubes all semester. Today we step back and notice something most textbooks never point out plainly: **every one of those formulas jumps up exactly one dimension.**

Look at the three formulas side by side:

$\sum_{k=1}^{n} k = 1 + 2 + \cdots + n$	$\frac{n(n+1)}{2}$	$\approx \frac{n^2}{2}$	**2D**
$\sum_{k=1}^{n} k^2 = 1^2 + 2^2 + \cdots + n^2$	$\frac{n(n+1)(2n+1)}{6}$	$\approx \frac{n^3}{3}$	**3D**
$\sum_{k=1}^{n} k^3 = 1^3 + 2^3 + \cdots + n^3$	$\left(\frac{n(n+1)}{2}\right)^2$	$\approx \frac{n^4}{4}$	**4D**

The Dimension-Jump Rule

Summing n terms of size k^d produces a result whose dominant term is n^{d+1}:

$$\boxed{\sum_{k=1}^{n} k^d \approx \frac{n^{d+1}}{d+1} \quad \text{as } n \to \infty}$$

One extra dimension appears every time you add everything up.

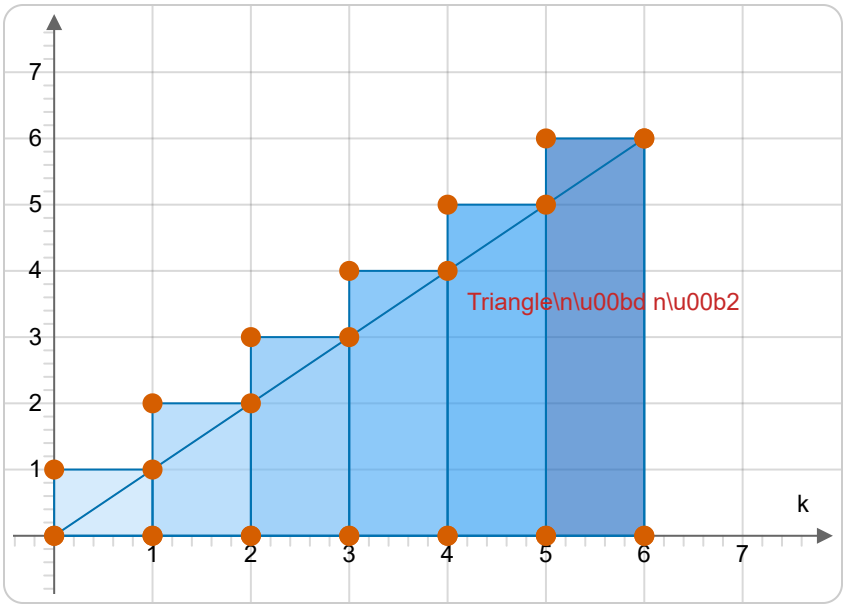
2. Geometric Story — Stacking Slabs

Why does adding 1D sticks produce 2D area? Picture it physically.

Visual A: $1 + 2 + \cdots + n$ as a Staircase

Draw n horizontal bars with lengths $1, 2, \ldots, n$ and stack them. You get a staircase triangle with base n and height n.

> Area of right triangle with legs n: $\frac{1}{2} n \cdot n = \frac{n^2}{2}$.
>
> **Exact formula $\frac{n(n+1)}{2}$ is slightly larger (the staircase "teeth"), but the dominant shape is the triangle.**

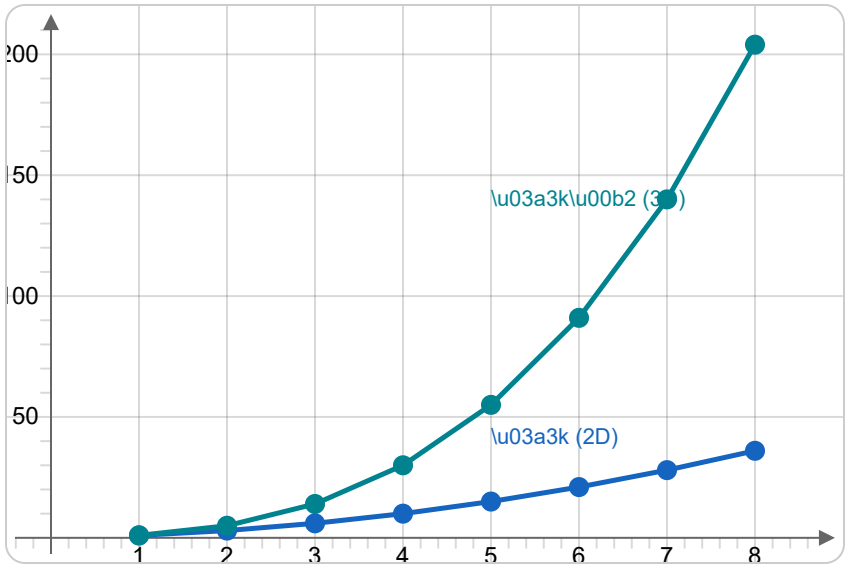


Staircase graph: bars of height 1, widths 1 through n (here $n = 6$). Gray triangle = $\frac{1}{2}n^2$ approximation.

Visual B: $1^2 + 2^2 + \cdots + n^2$ as Stacking Squares

Each term k^2 is a $k \times k$ square (2D slab of thickness 1). Stack them → stepped pyramid.

> Pyramid with base $n \times n$ and height n: volume $\approx \frac{1}{3} n^2 \cdot n = \frac{n^3}{3}$.
>
> **Exact formula $\frac{n(n+1)(2n+1)}{6}$ has leading term $\frac{n^3}{3}$. Adding 2D slabs builds a 3D shape.**



Comparing $y = \sum_{k=1}^{x} k$ *(blue,* $\approx x^2/2$*) vs* $y = \sum_{k=1}^{x} k^2$ *(teal,* $\approx x^3/3$*) for* $x = 1 \ldots 8$.

"Add d-dimensional objects together → you fill a $(d+1)$*-dimensional container."*

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3. The Cube-Sum Magic: $\sum k^3 = \left(\sum k\right)^2$

Nicomachus's Theorem (c. 100 AD)

$$1^3 + 2^3 + \cdots + n^3 \;=\; (1 + 2 + \cdots + n)^2 \;=\; \left[\frac{n(n+1)}{2}\right]^2$$

The sum of the first n cubes equals the *square* of the sum of the first n integers. This works **only** for cubes.

Quick Check: $n = 4$

Left: $1^3 + 2^3 + 3^3 + 4^3 = 1 + 8 + 27 + 64 = 100.$

Right: $\left(\frac{4 \cdot 5}{2}\right)^2 = 10^2 = 100.$ ✓

Nicomachus verified for $n = 4$**.**

Leading term: $\frac{n^4}{4}$ → firmly 4D, consistent with the pattern.

4. The General Pattern

Power-Sum Approximation (all positive integers d)

$$\sum_{k=1}^{n} k^d \;=\; \frac{n^{d+1}}{d+1} \;+\; \text{lower-order terms}$$

As n grows large, lower-order terms become negligible. The dominant behavior is always $\dfrac{n^{d+1}}{d+1}$.

The exact formula for general d uses **Bernoulli numbers** (Number Theory / Calculus topic), but the approximation above is the key takeaway.

5. This Is Secretly Calculus

The integral of x^d from 0 to n:

$$\int_0^n x^d\,dx \;=\; \left.\frac{x^{d+1}}{d+1}\right|_0^n \;=\; \frac{n^{d+1}}{d+1}$$

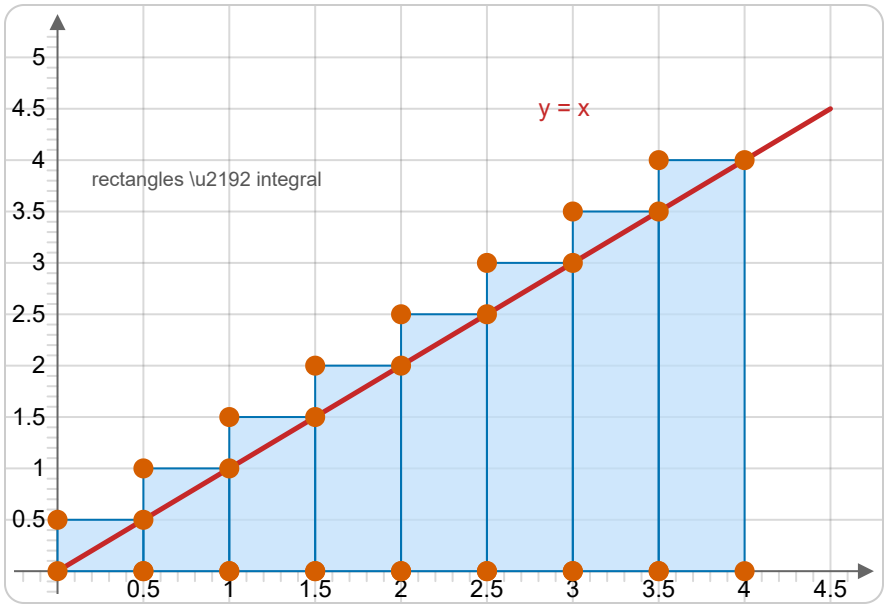
Compare:

$$\sum_{k=1}^{n} k^d \;\approx\; \frac{n^{d+1}}{d+1}$$

Discrete Sums = Continuous Integrals (in the limit)

The Riemann integral is literally defined as the limit of sums like these. When you let $n \to \infty$ (shrinking each step to width $\frac{1}{n}$), the sum *becomes* the integral.

Punchline: The antiderivative rule $\displaystyle\int x^d\,dx = \frac{x^{d+1}}{d+1}$ is the continuous version of the dimension-jump pattern in finite sums. *Pre-Calculus was previewing Calculus the whole time.*



Riemann sum (rectangles, $d = 1$, $n = 8$) vs $y = x$ on $[0, 4]$. As rectangles shrink, sum $\to$ integral $= \frac{1}{2}(4)^2 = 8$.

6. Induction Proof of Row 1

The two-step induction machine proves the dimension-jump formula for Row 1. Here is the complete proof as a model.

Theorem: $\displaystyle\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$

Proof by Mathematical Induction.

Base Case ($n = 1$): LHS $= 1$. RHS $= \dfrac{1 \cdot 2}{2} = 1$. ✓

Inductive Hypothesis ($n = m$): Assume $\displaystyle\sum_{j=1}^{m} j = \frac{m(m+1)}{2}$ for some $m \geq 1$.

Inductive Step ($n = m + 1$):

$$\sum_{j=1}^{m+1} j = \underbrace{\sum_{j=1}^{m} j}_{\frac{m(m+1)}{2}} + (m+1) = \frac{m(m+1)}{2} + (m+1) = (m+1)\left(\frac{m}{2} + 1\right) = \frac{(m+1)(m+2)}{2}$$

This is the formula with $n = m + 1$. By PMI, the formula holds for all $n \geq 1$. ■

Dimension in the Proof

At the inductive step we added a 1D term $(m + 1)$ to a formula that was already quadratic (2D). The result stayed quadratic. Induction and geometry are telling the same story.

7. Summary — The Three Pillars

1D Sums	$\frac{n(n+1)}{2} \approx \frac{n^2}{2}$	Sticks → triangle (area, 2D)
2D Sums	$\frac{n(n+1)(2n+1)}{6} \approx \frac{n^3}{3}$	Squares → pyramid (volume, 3D)
3D Sums	$\left[\frac{n(n+1)}{2}\right]^2 \approx \frac{n^4}{4}$	Cubes → 4D hyper-volume; $= (\sum k)^2$
Calculus limit	$\int_0^n x^d \, dx = \frac{n^{d+1}}{d+1}$	Continuous version of the same jump

One-Line Takeaway

Summing d-dimensional things builds a $(d+1)$-dimensional thing — whether discrete (sigma) or continuous (integral), the rule is always $\frac{n^{d+1}}{d+1}$.

Practice Problems

Directions: *Show all work. Leave answers in exact form unless otherwise stated. Problems 1–4: Pattern Recognition | Problems 5–7: Nicomachus & Algebra | Problems 8–10: Calculus Connection & Induction*

Part A — Pattern Recognition

1. Fill in the table below. For each sum, identify the leading term and the dimension of the result.

$\sum_{k=1}^{n} k$			
$\sum_{k=1}^{n} k^2$			
$\sum_{k=1}^{n} k^3$			

2. Using the dimension-jump rule $\sum_{k=1}^{n} k^d \approx \frac{n^{d+1}}{d+1}$, predict the leading term of $\sum_{k=1}^{n} k^4$. What dimension does the result live in?

3. Compute $\sum_{k=1}^{10} k$ and $\sum_{k=1}^{10} k^2$ using the exact formulas. Then compare their ratio to the ratio of their leading terms. Are they close?

4. A student claims: "The sum of the first n fourth-powers has leading term $\frac{n^5}{5}$." Verify this claim by checking the formula at $n = 2$ and $n = 3$ using

$$1^4 + 2^4 + \cdots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

Part B — Nicomachus & Algebraic Verification

5. Verify Nicomachus's Theorem for $n = 5$: show that $1^3 + 2^3 + 3^3 + 4^3 + 5^3 = \left(\frac{5 \cdot 6}{2}\right)^2$.

6. Using Nicomachus's Theorem, simplify and evaluate:

$$\frac{1^3 + 2^3 + 3^3 + \cdots + n^3}{(1 + 2 + 3 + \cdots + n)^2}$$

What is the value of this expression for any $n \geq 1$? Explain why.

7. Find the value of n such that $1^3 + 2^3 + \cdots + n^3 = 3025$. *(Hint: use Nicomachus's Theorem and recognize a perfect square.)*

Part C — Calculus Connection & Induction

8. Evaluate the following integral and compare your answer to the sum formula:

$$\int_0^{10} x^2\,dx$$

Then compute $\displaystyle\sum_{k=1}^{10} k^2$ exactly. Express the ratio $\dfrac{\sum_{k=1}^{10} k^2}{\int_0^{10} x^2\,dx}$ as a decimal rounded to 3 places. What do you notice as n grows large?

9. The antiderivative rule says $\displaystyle\int x^3\,dx = \frac{x^4}{4} + C$. Explain in one or two sentences how this rule is the "continuous version" of the dimension-jump pattern $\displaystyle\sum_{k=1}^{n} k^3 \approx \frac{n^4}{4}$.

10. Prove by mathematical induction that $\displaystyle\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$. Follow the three-step structure: **(a)** Base case $n = 1$. **(b)** State the inductive hypothesis. **(c)** Prove the inductive step.